

Computer Organisation

Topic 10 Memory

Stallings, Computer Organization & Architecture (8th Ed.), Chap. 5

Hamacher et al., Computer Organisation (5th Ed.), Chap. 5

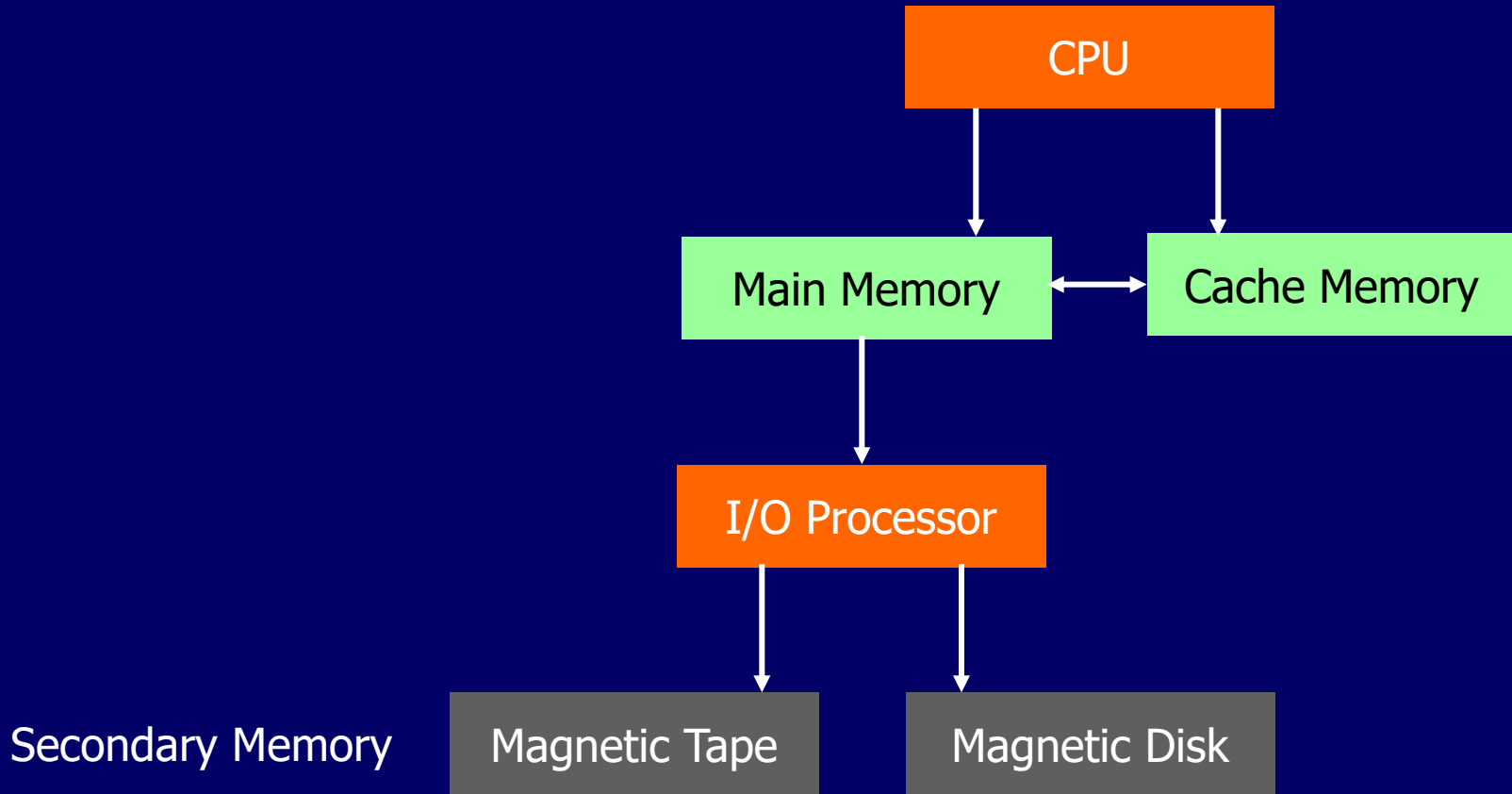
Outline

- Computer Memory Overview
 - Memory hierarchy
- Semiconductor Main Memory
 - RAM/ROM
 - Organisation of Memory Chip
- Error Correction
- Advanced DRAM Organisation

Memory

- For each classification there are several types of memories that can be differentiated by its making
- 2 classifications
 - Main Memory
 - Secondary Memory

Memory



Characteristics

- Location
- Capacity
- Unit of transfer
- Access method
- Performance
- Physical type
- Physical characteristics
- Organisation

Location

- CPU
 - Registers
- Internal
 - Cache
 - Main memory
- External
 - Peripheral storage
 - Secondary memory

Capacity

- Expressed in terms of bytes (1 byte = 8 bits) or words
- Word size
 - The natural unit of organisation
 - Word lengths are 8, 16, and 32 bits
- Number of words
 - Or Bytes

Unit of Transfer

- Internal
 - Usually governed by data bus width into and out of the memory module.
 - Normally equal to word length, but can be larger, such as 64, 128 or 256 bits
- External
 - Usually a block which is much larger than a word

Unit of Transfer

- Related concepts:
 - Word
 - Word size typically equal to number of bits used to represent a number/instruction length
 - Addressable unit
 - Smallest location which can be uniquely addressed
 - Word internally
 - Relation between the length (in bits) A of an address and the number N of addressable units is $2^A = N$
 - Unit of transfer
 - Number of bits read/written into memory at a time
 - Need not equal a word/addressable unit but can be larger, e.g. blocks (for external memory)

Access Methods (1)

- Sequential
 - Memory is organized into unit of data, called records
 - Start at the beginning and read through in order (specific linear sequence)
 - Access time depends on location of data and previous location
 - e.g. tape
- Direct
 - Individual blocks have unique address
 - Access is by jumping to vicinity plus sequential search
 - Access time depends on location and previous location
 - e.g. disk

Access Methods (2)

- Random
 - Individual addresses identify locations exactly
 - Access time is independent of location or previous access
 - No “searching” required
 - e.g. RAM
- Associative
 - Data is located by a comparison with contents of a portion of the store
 - Word is retrieved based on a portion of content rather than on address
 - Access time is independent of location or previous access
 - e.g. cache

Performance (1)

- Access time
 - For RAM, time between presenting the address and getting the valid data
 - For non RAM, the time it takes to position the read-write mechanism at the desired location
- Memory Cycle time
 - Primarily applied to RAM
 - Time may be required for the memory to “recover” before next access
 - Access time + recovery
 - For transients to die out on signal lines
 - Concerned with the system bus not the processor

Performance (2)

- Transfer Rate
 - Rate at which data can be moved in and out of memory unit
 - For RAM = $1/(\text{cycle time})$
 - Non-RAM: $T_N = T_A + N/R$ (i.e. access + transfer)
 - T_N = average time to read or write N bits
 - T_A = average access time
 - N = number of bits
 - R = transfer rate, in bits per second (bps)

Physical Types

- Semiconductor
 - RAM
- Magnetic
 - Disk & Tape
- Optical
 - CD & DVD
- Others
 - Hologram

Physical Characteristics

- Volatile
 - Information lost when power is switched off
 - E.g. RAM
- Non-volatile
 - Information remains until deliberately changed
 - No power needed to maintain information
 - E.g. magnetic surface memories
- Non-erasable
 - ROM

Organisation

- Physical arrangement of bits into words
- e.g. interleaved

Design Constraints (1)

- How much the capacity?
 - Capacity
- How fast the access time?
 - The memory must be able to keep up with the processor
- How expensive?
 - Must be reasonable in relationship to other components
- Memory trade-off
 - Cost, capacity, and access time

Design Constraints (2)

- The following relationships hold
 - Faster access time, greater cost per bit
 - Greater capacity, smaller cost per bit
 - Greater capacity, slower access time

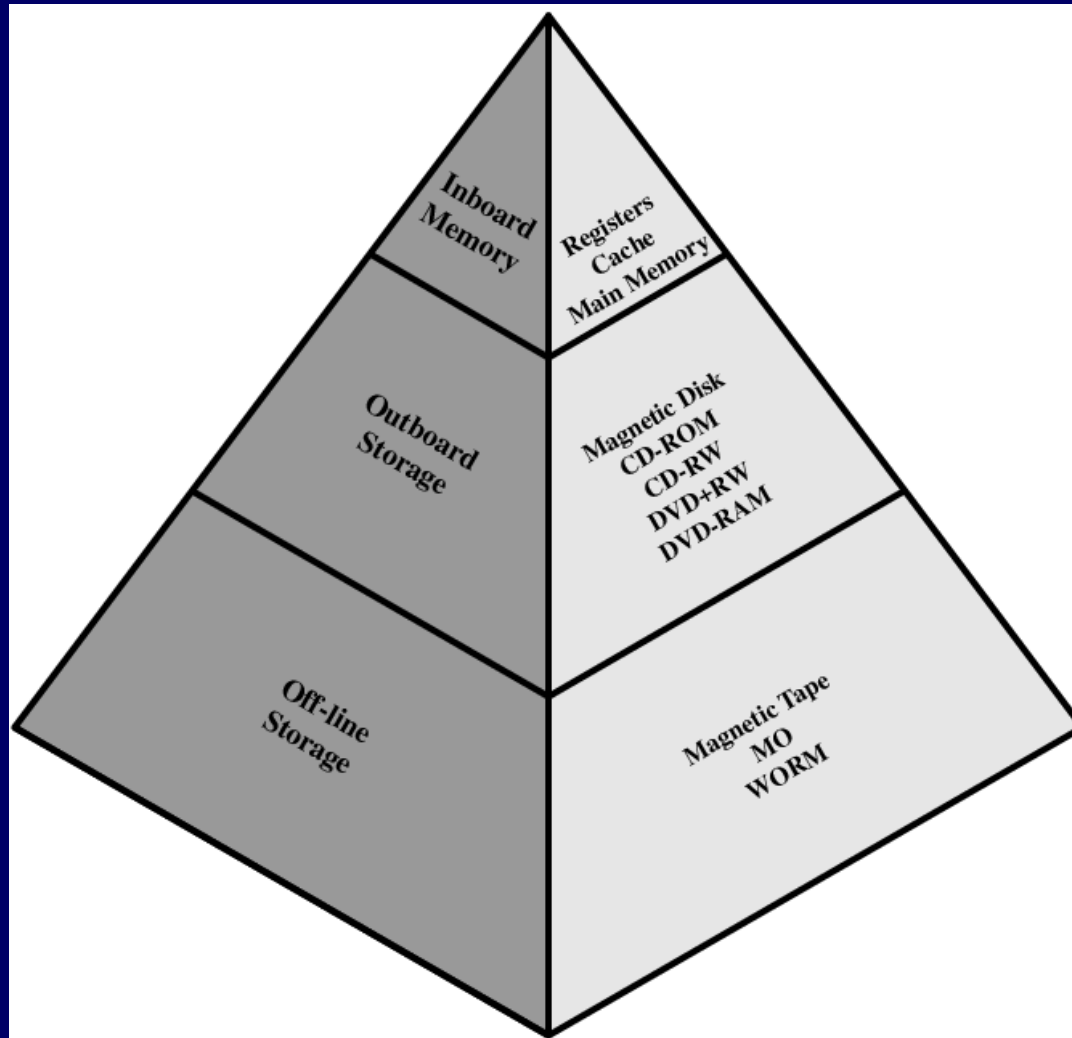
Design Solution

- Not to rely on single memory component/technology
- But employ a memory hierarchy

Memory Hierarchy

- Registers
 - In CPU
- Internal or main memory
 - May include one or more levels of cache
 - RAM
- External memory
 - Backing store

Memory Hierarchy - Diagram



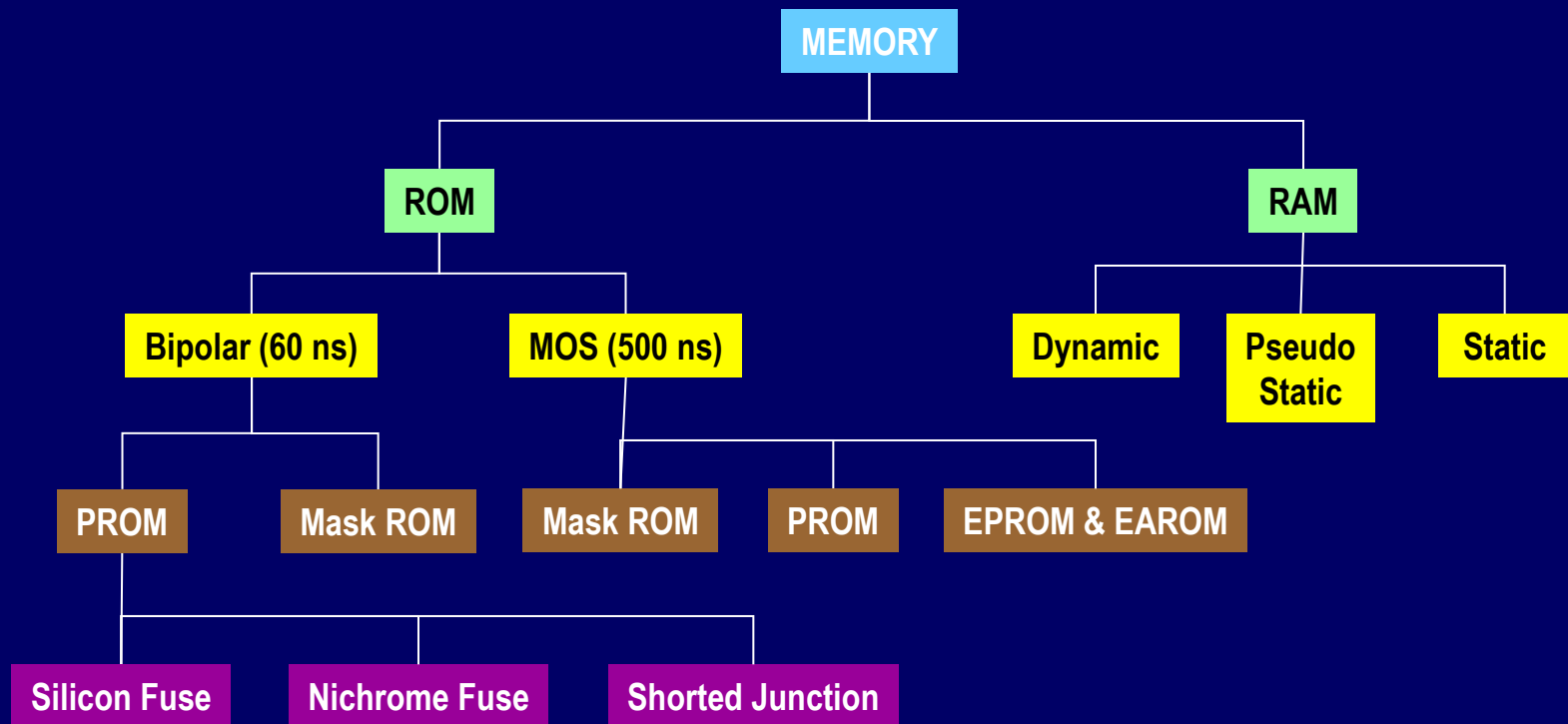
Hierarchy List

- Registers
 - L1 Cache
 - L2 Cache
 - Main memory
 - Disk cache
 - Disk
 - Optical
 - Tape
- 
- Decreasing cost per bit
 - Increasing capacity
 - Increasing access time
 - Decreasing frequency of memory access by processor

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Main Memory

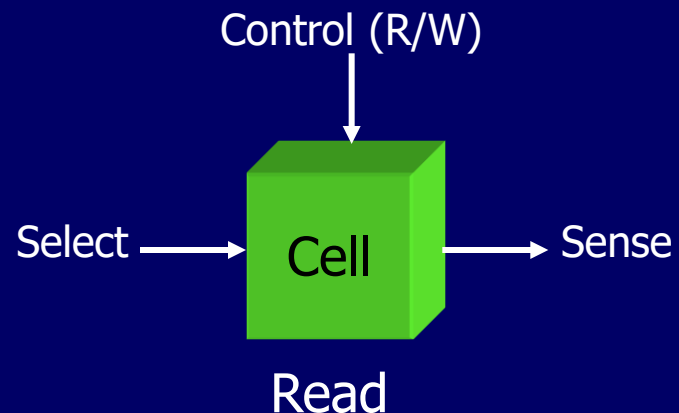
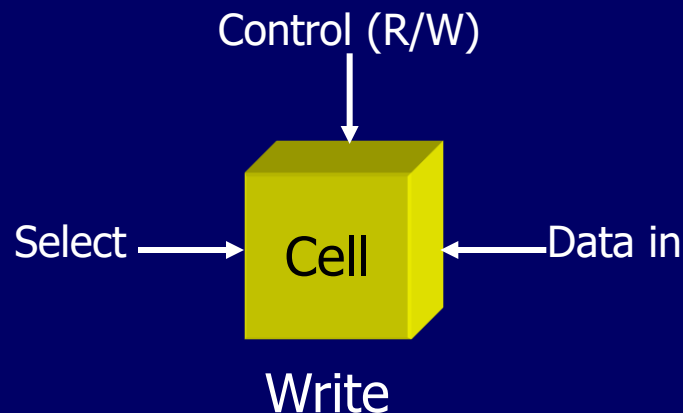


Main Memory

- Main memory implementation using integrated circuit
- Main memory stores program and data
- Computer speed = speed of data transfer from main memory
- Physical size of main memory is usually not as big as the specified address by program
 - Only blocks that are needed are stored in main memory
 - Others - secondary memory
- Maximum size of main memory depends on the addressing scheme

Semiconductor Main Memory

- Basic element is memory cell
 - 2 stable states
 - Represent binary 1 and 0
 - Capable of being written into (at least once) to set the state
 - Capable of being read to sense the state



Random Access Memory (RAM)

- Memory unit that can be accessed for read and write operation
- 2 types of RAM
 - Static RAM (SRAM)
 - Dynamic RAM (DRAM)

SRAM

- Static RAM
- Is a digital device, using the same logic elements used in the processor
- Binary values are stored using traditional flip-flop logic gate
- Hold its data as long as power is supplied to it

SRAM



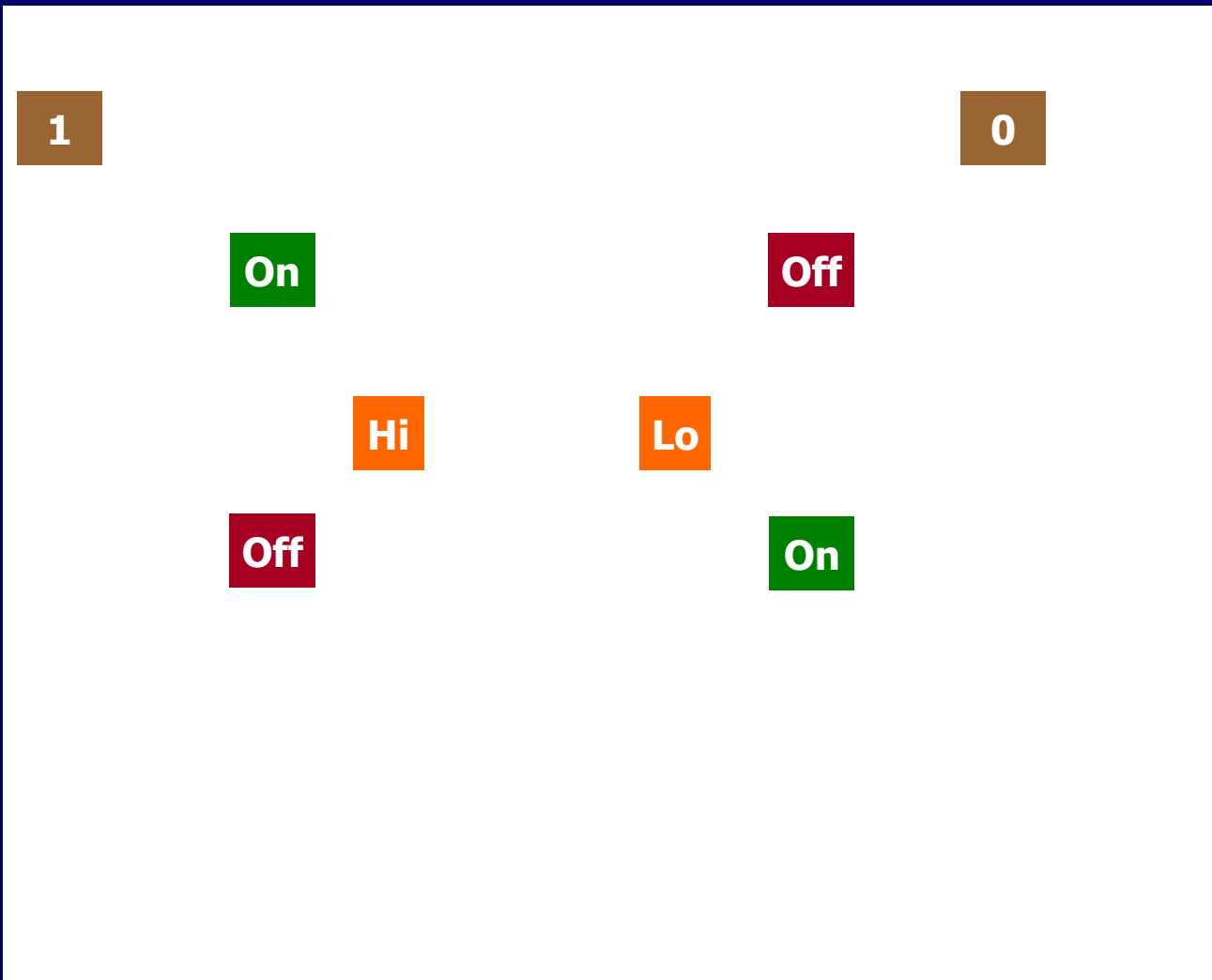
SRAM

- For state 1
 - Voltage at X is high, at Y is low
 - T_3 and T_6 are on (closed)
 - T_4 and T_5 are off (open)
- For state 0
 - Voltage at X is low, at Y is high
 - T_4 and T_5 are on (closed)
 - T_3 and T_6 are off (open)

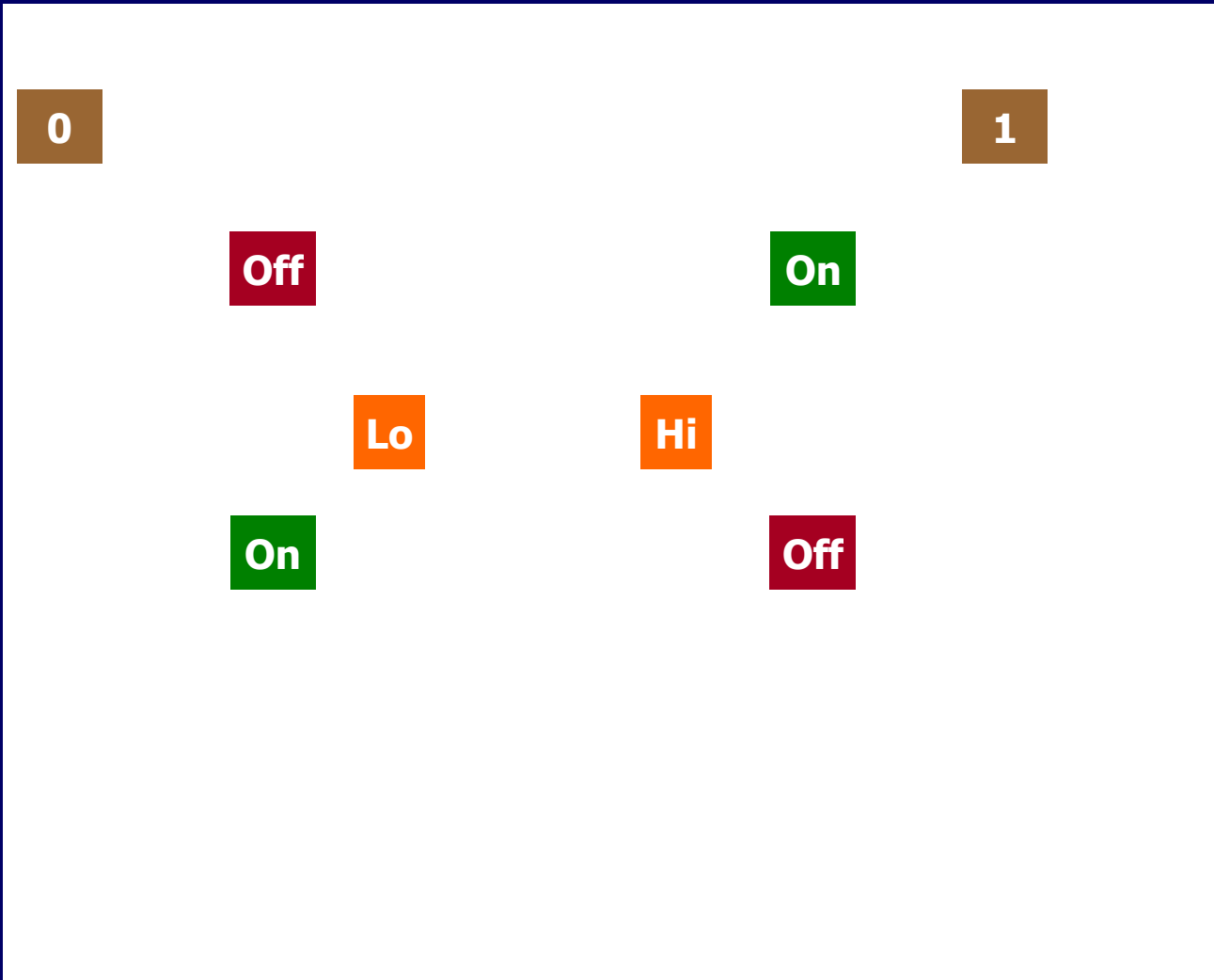
SRAM

- Address/word line used to open/close a switch
 - Address/word line controls T_1 and T_2
 - When signal is applied, T_1 and T_2 are switched on, allowing read/write
- Write
 - Desired bit value is applied to line b , while its complement to line b'
 - This forces T_1 , T_2 , T_3 , and T_4 into the proper state
- Read
 - Value is read from line b

SRAM (State 1)



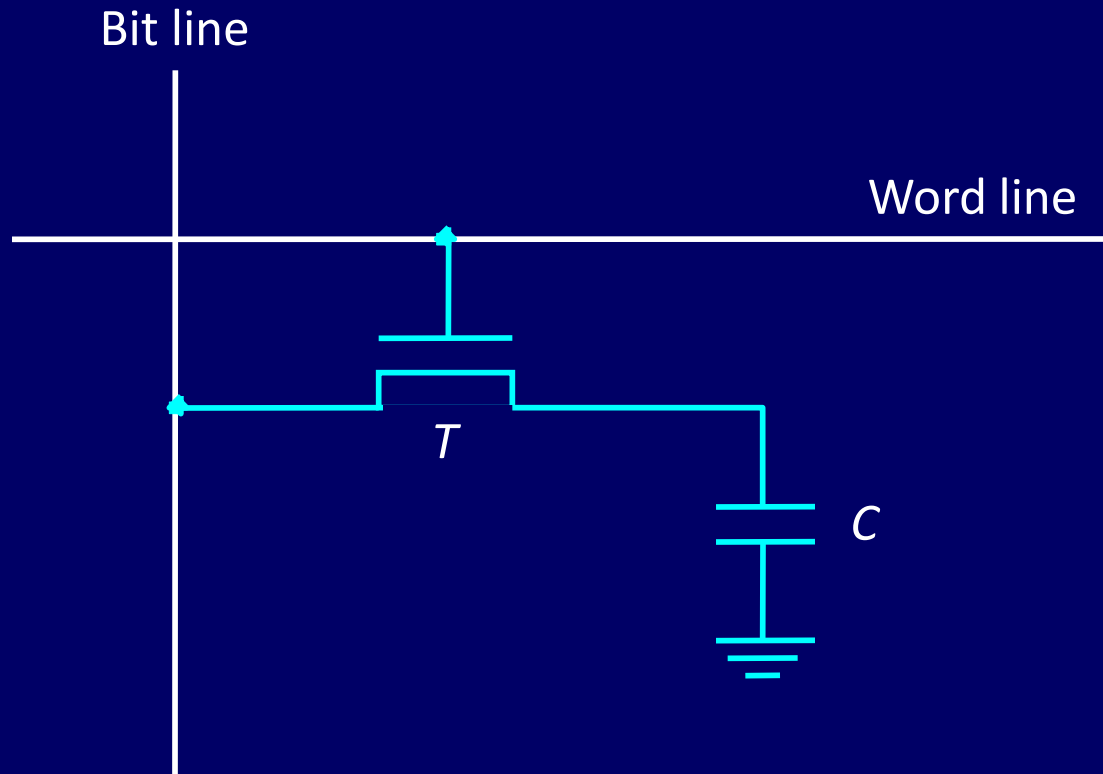
SRAM (State 0)



DRAM

- Dynamic RAM
- Made with cells that store data as charge on capacitors
 - Charge maintained for only tens of milliseconds
 - Requires periodic charge to maintain data storage
- The absent and presence of the charge is interpreted as binary 0 and 1

DRAM

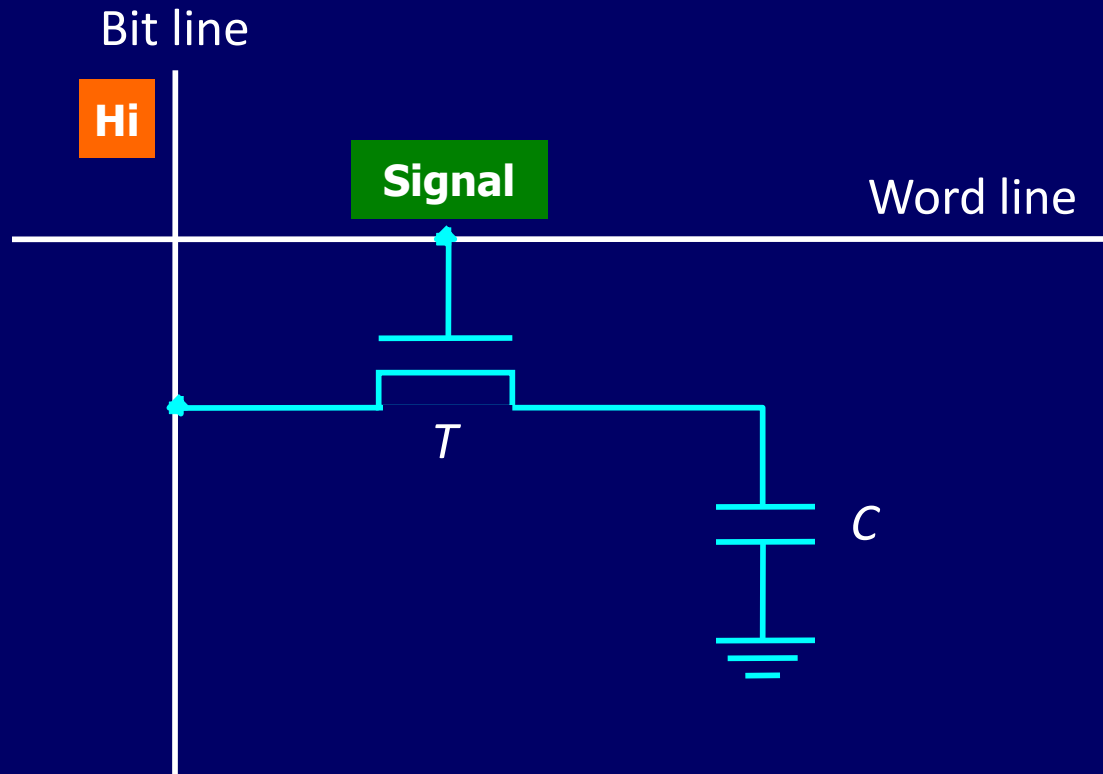


A single-transistor dynamic memory cell

DRAM

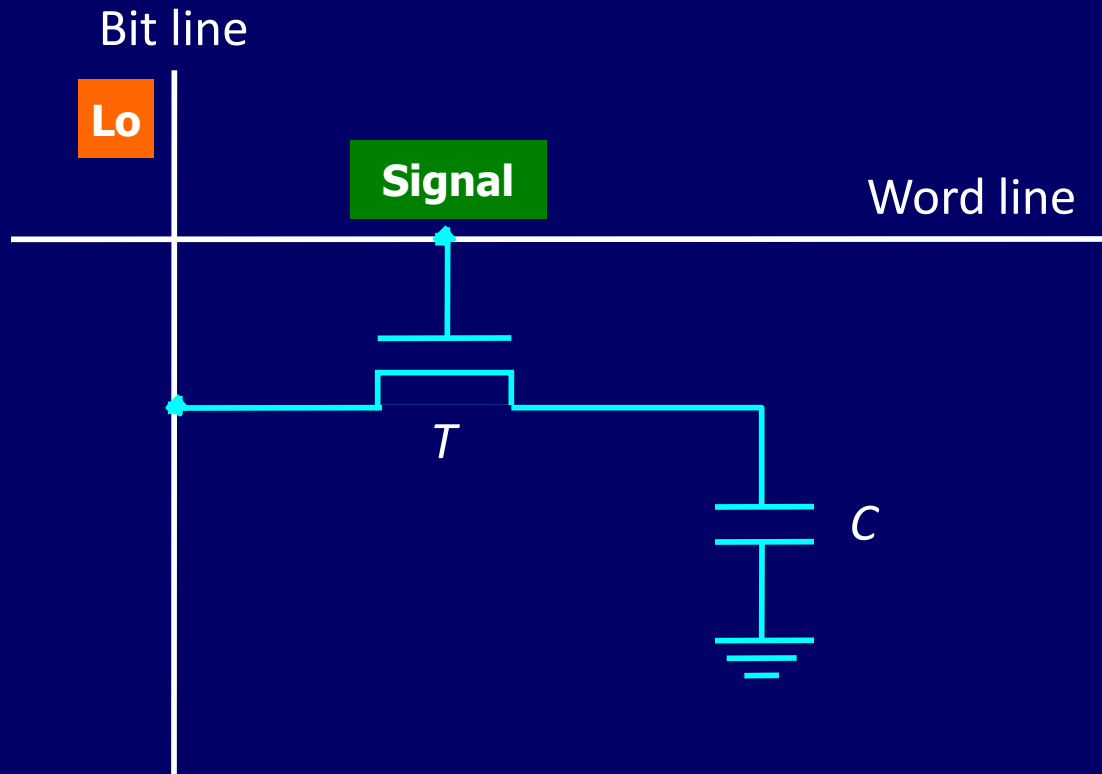
- Write
 - Voltage signal applied to bit line (high voltage = 1; low voltage = 0)
 - Signal applied to address/word line
 - Charge transferred to capacitor
- Read
 - Address/word line selected
 - Transistor turns on (closed/allow current)
 - Charge stored on capacitor fed out onto bit line
 - Sense amplifier compares capacitor voltage to reference value (threshold) to determine if value is 1 or 0
 - Read discharges capacitor – need to recharge to complete read operation

DRAM (Write 1)



A single-transistor dynamic memory cell

DRAM (Write 0)



A single-transistor dynamic memory cell

SRAM vs DRAM

- SRAM and DRAM are volatile
 - Power must be continuously supplied to the memory
- DRAM cell is simpler and smaller than a SRAM cell
- DRAM is more dense and less expensive than SRAM
- DRAM requires the supporting refresh circuit
- DRAM is used for large memory requirements
- SRAM is faster than DRAM
- SRAM is used for cache memory, DRAM is used for main memory

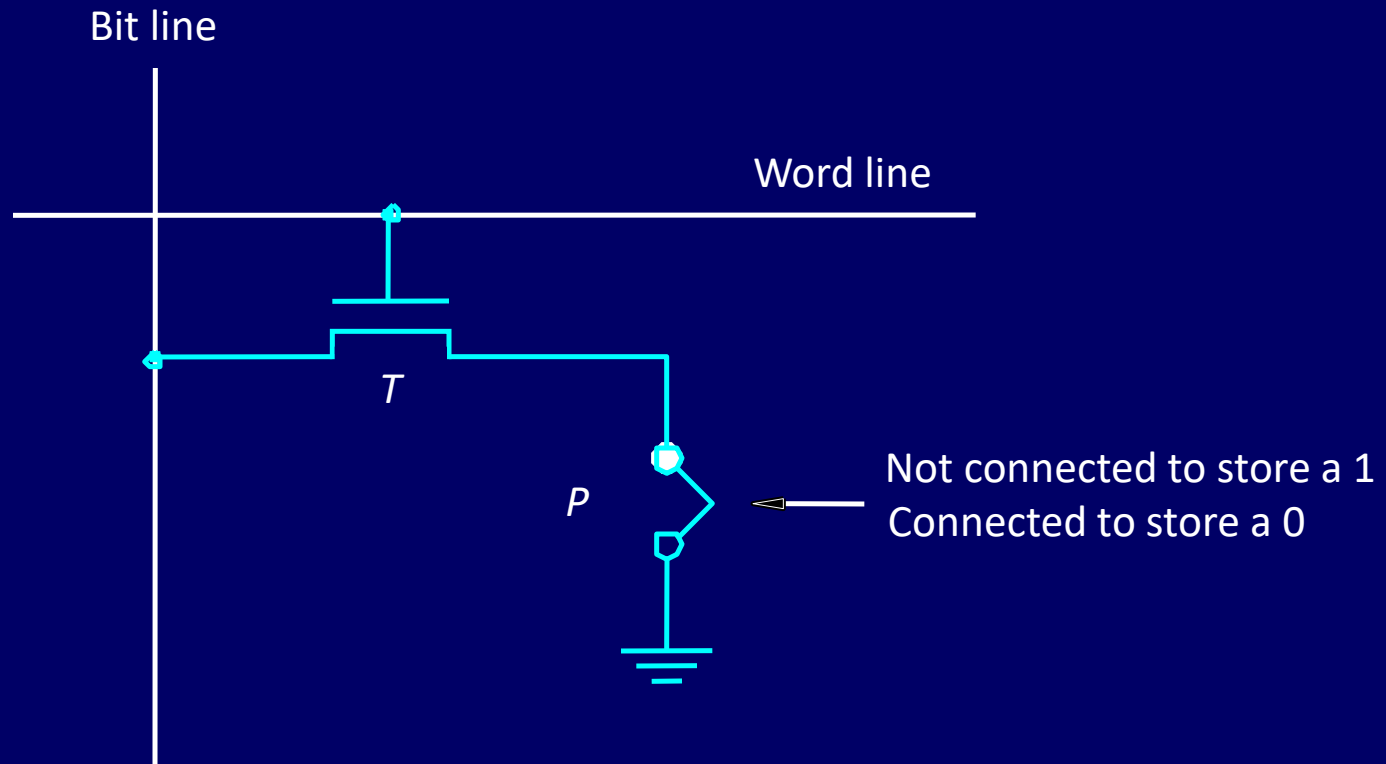
ROM

- Permanent storage
 - Nonvolatile - no power source is required to maintain the bit values in memory
- Can read, cannot write new data into it
- Application
 - microprogramming, library subroutine, system programs (BIOS) and function tables
- Advantage
 - Data and program are permanent in main memory and no need to load from secondary memory

ROM

- Data actually wired into the chip during fabrication
 - Disadvantage
 - Data insertion involves large fixed cost
 - No room for error

ROM



Types of ROM

- Programmable (once)
 - PROM
 - Needs special equipment to program
 - Programming done after fabrication (electronically)
 - Before programming, all are 0s
 - Value 1 is achieved by burning the fuse at P
 - Flexible and convenient
- Read Mostly Memory
 - Erasable Programmable (EPROM)
 - Electrically Erasable (EEPROM)
 - Flash memory

Types of ROM

- Erasable Programmable (EPROM)
 - Read and written electronically
 - Special transistor is used at P
 - Before writing, all storage cells must be erased to same initial state
 - Erased by UV light
- Electrically Erasable (EEPROM)
 - No need to erase prior content – only selected byte(s) are erased
 - Takes much longer to write than read
 - More expensive than EPROM and less dense

Types of ROM

- Flash memory
 - Intermediate between EPROM/EEPROM in cost and functionality
 - Erase whole memory electrically, faster erasure than EPROM
 - Possible to erase blocks of memory instead of entire chip
 - But does not provide byte-level erasure
 - High density – like EPROM

Internal Organisation of Memory Chip

- Memory cells organised in the form of a string or an array
- Each cell = 1 bit information
- Each line of cell = 1 word
- Each line of cell is connected with word line - $W_0, W_1, \dots$ to address decoder

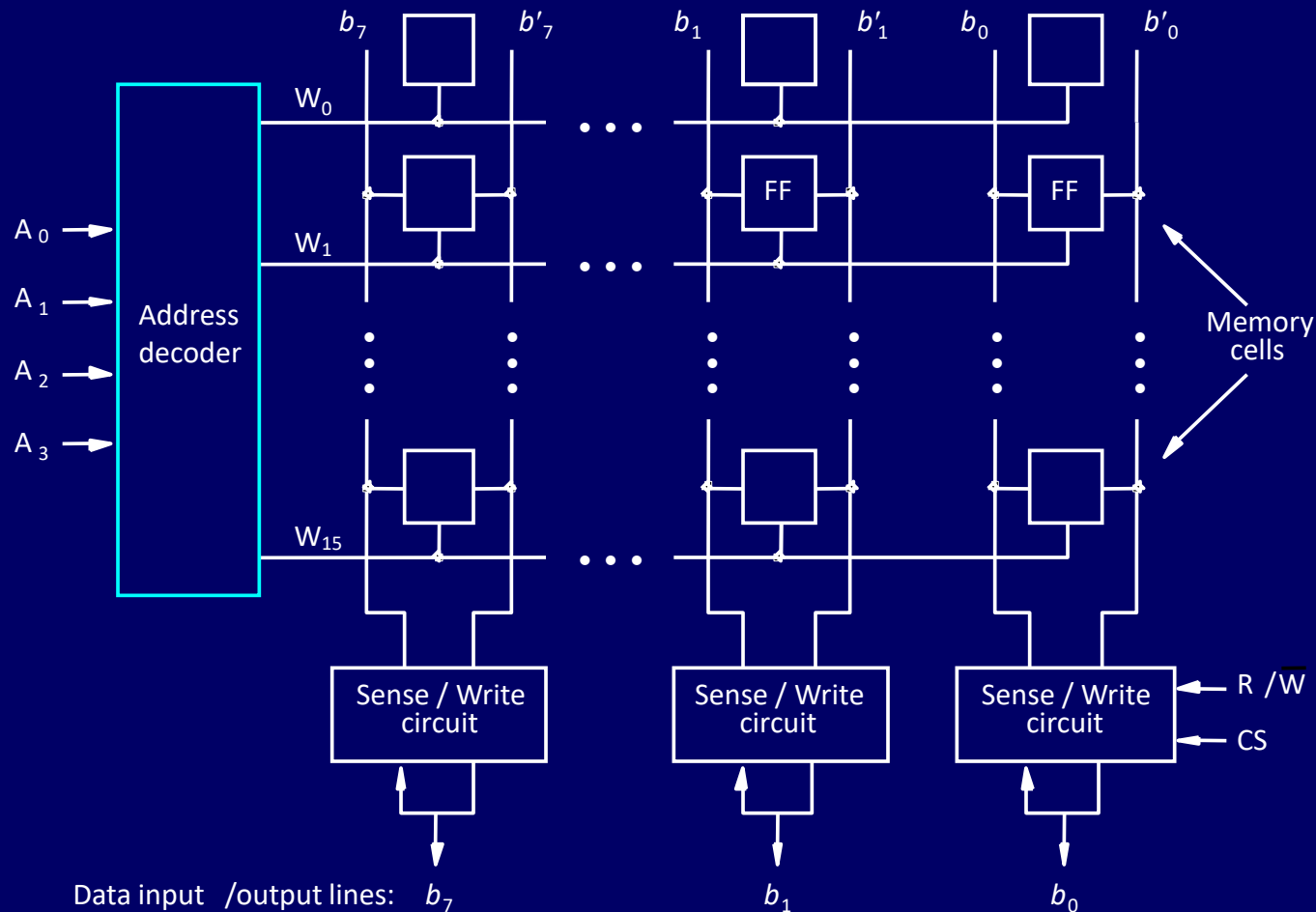
Internal Organisation of Memory Chip

- Each cell has sense/write circuit
 - Read operation
 - Sense/write circuit reads content through data line
 - Write operation
 - Circuit receives data and stores it in selected word cell
- Example 16 x 8 organisation
 - 16 words, 8 bits
- R/W and CS line
 - R/W specifies operation
 - CS (chip select) selects chip in a multi-chip system

Internal Organisation of Memory Chip

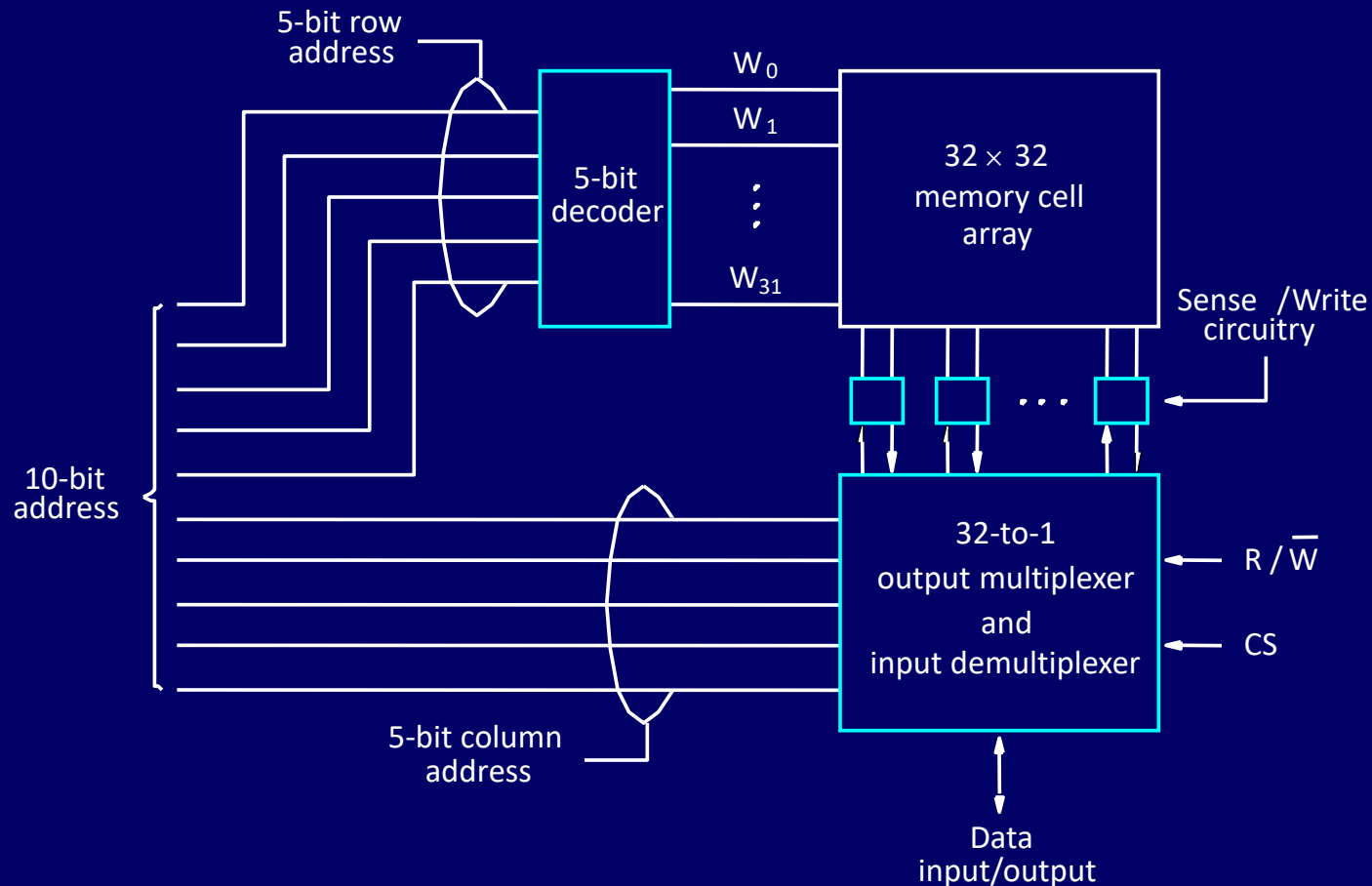
- Stores 128 bit
- 14 external connections
 - 4 for word, 8 for bit, 2 for R/W+CS
 - Can be built as 16 pin chip
 - 14 + 2 for power supply and ground
- How many pins required for 1K (1024 bit) memory cell?
 - 128 x 8 7 for word, 8 for bit, 2 for R/W+CS = 17+2 pins
 - 1024 x 1 10 for word, 1 for bit, 2 for R/W+CS = 13+2 pins

Internal Organisation of Memory Chip



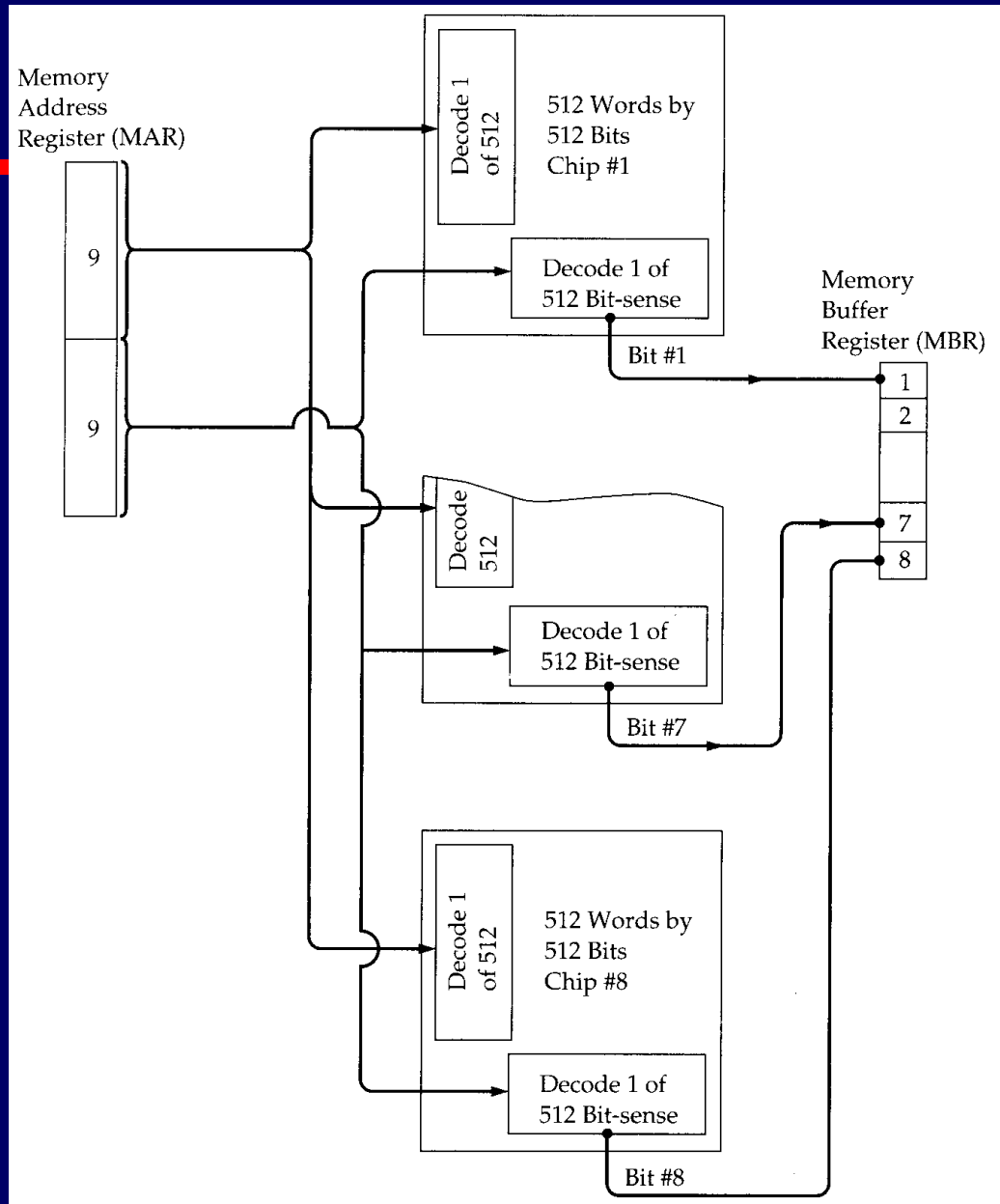
Organization of bit cells in a memory chip (128 bit)

Internal Organisation of Memory Chip



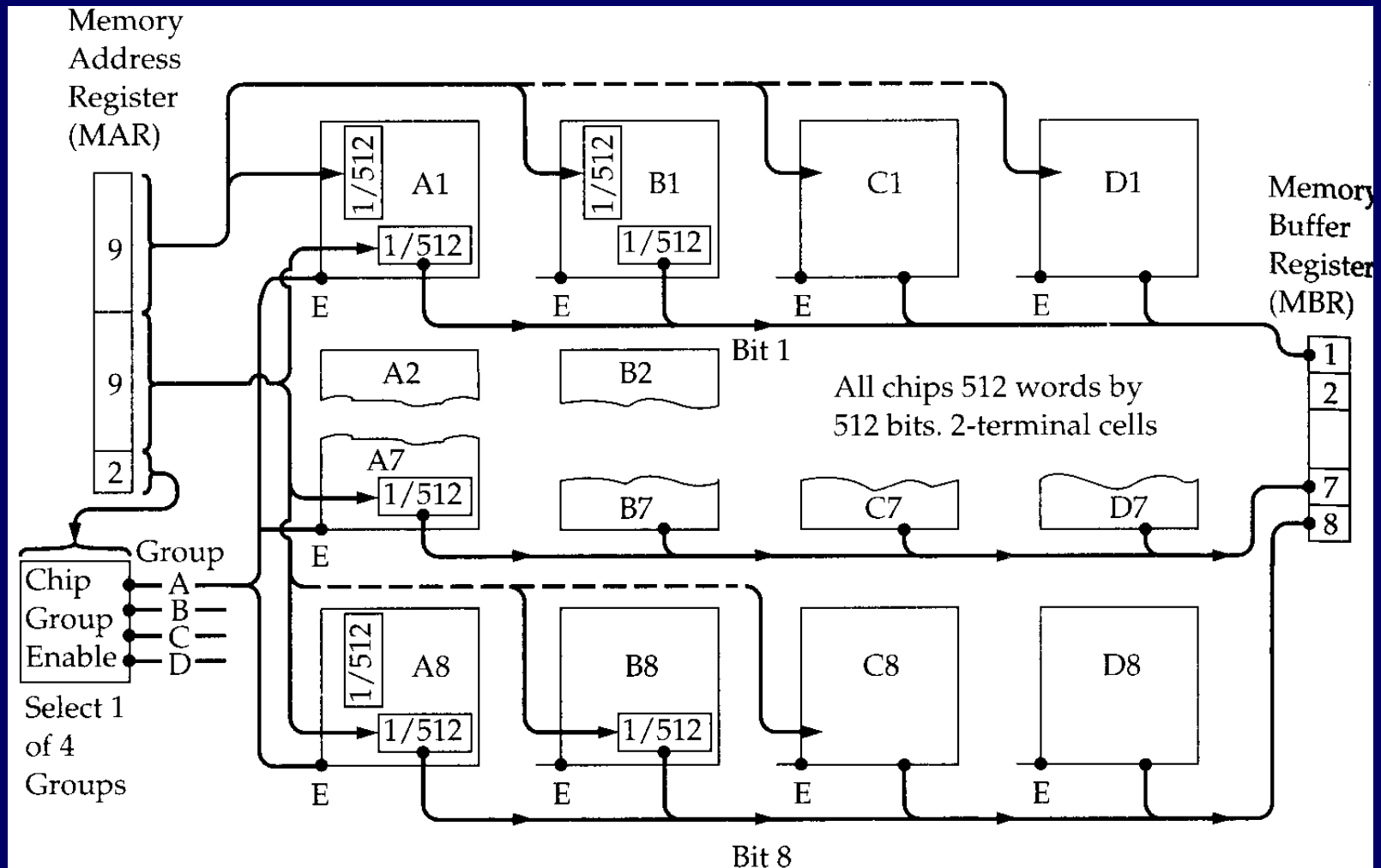
Organization of a 1K x 1 memory chip

Internal Organisation of Memory Chip



256K 8-bit words = 2M
Use 8 512 words x 512 bit chips
Each chip produces 1 bit out of the 8 bit word
To address 256K words, 18 bits required

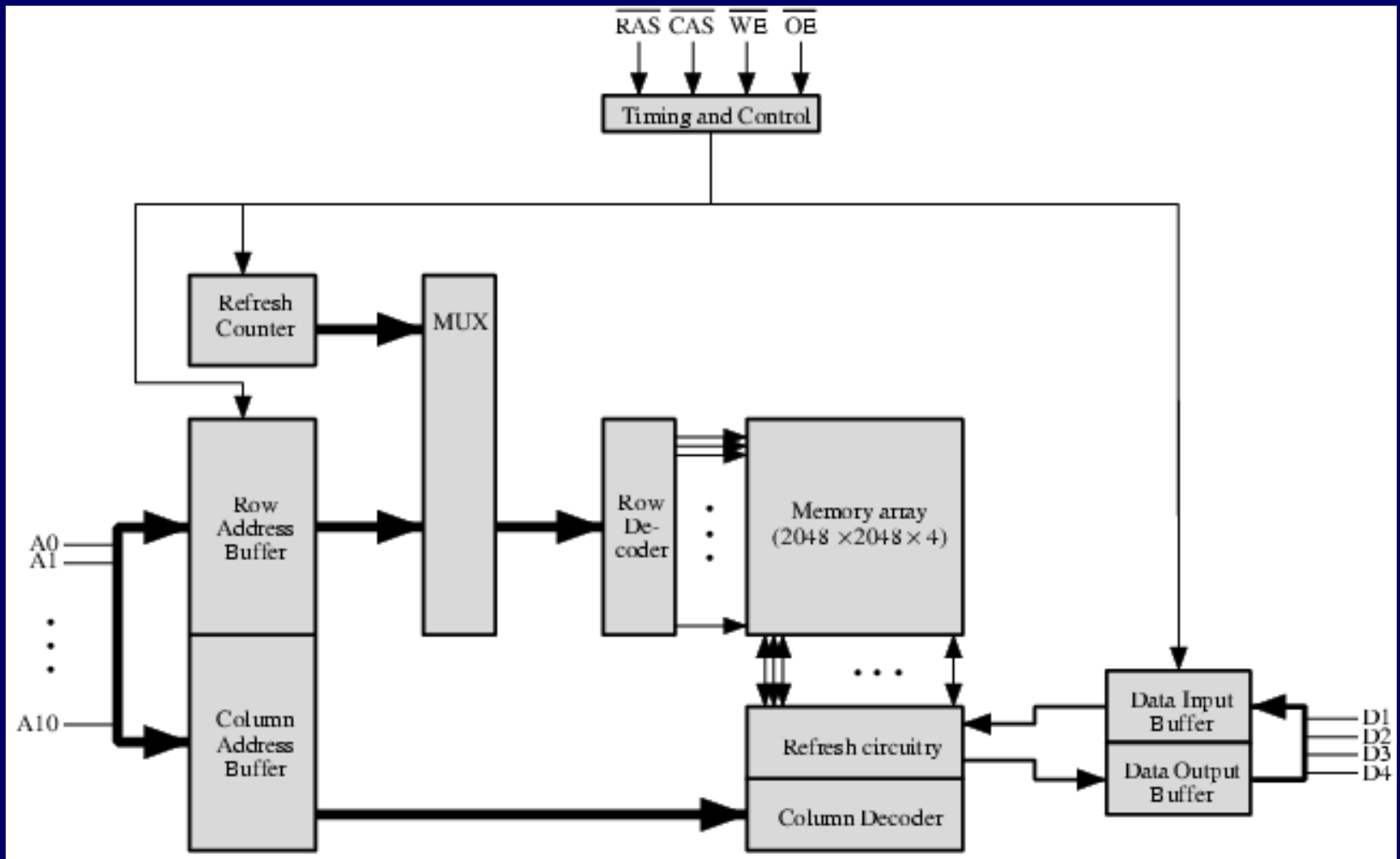
Internal Organisation of Memory Chip



Another Example (16 MBit)

- A 16Mbit chip can be organised as 1M of 16 bit words
- A bit per chip system has 16 lots of 1Mbit chip with bit 1 of each word in chip 1 and so on
- A 16Mbit chip can be organised as a 2048 x 2048 x 4bit array
 - Reduces number of address pins
 - Multiplex row address and column address
 - 11 pins to address ($2^{11}=2048$)
 - Adding one more pin doubles range of values so x4 capacity

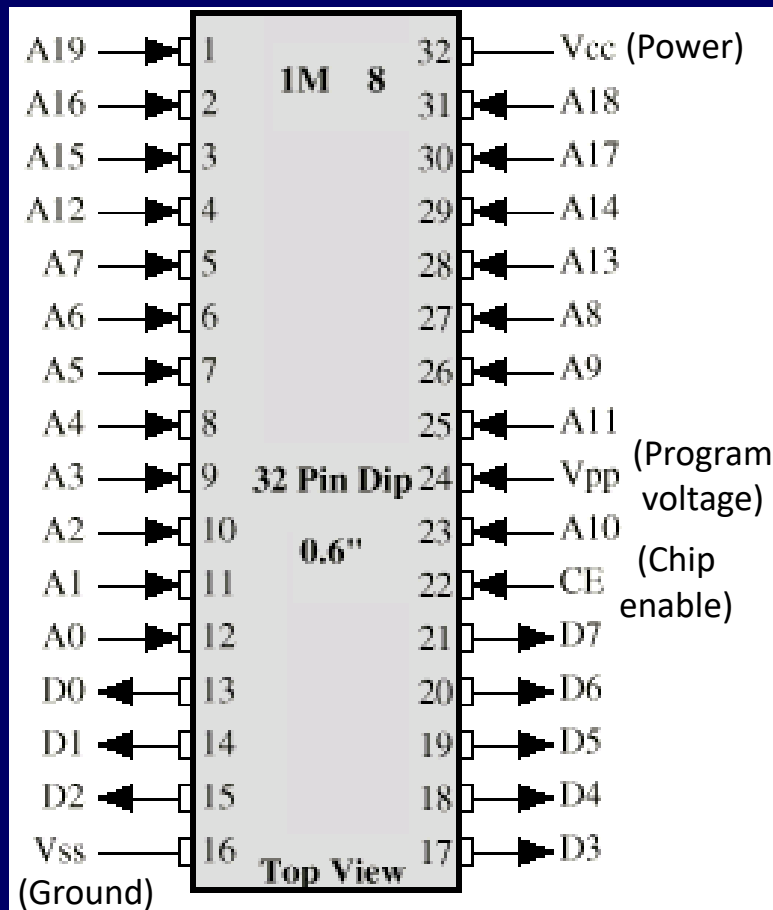
Typical 16 Mb DRAM (4M x 4)



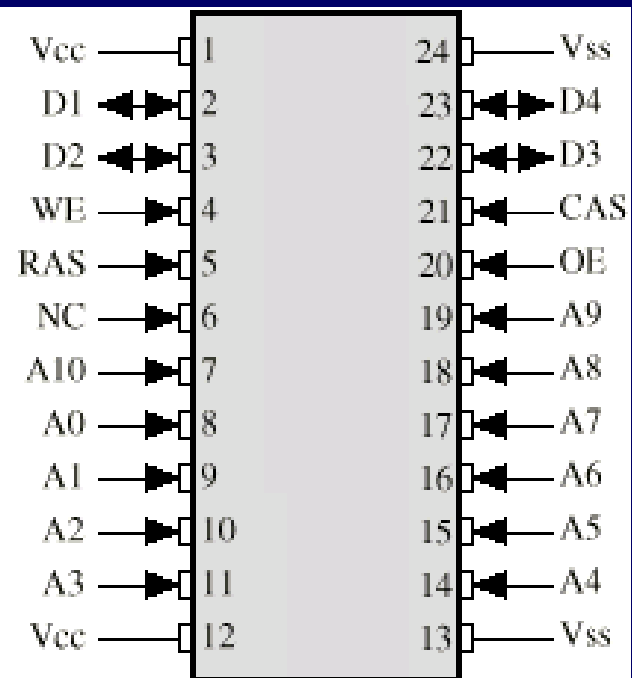
Refresh Circuit

- All DRAMS require refreshing
- Refresh circuit included on chip
- Disable chip while cells are refreshed
- Refresh counter steps through all rows
 - For each row, output lines supplied to row decoder
 - Data read out and written back
- Takes time
- Slows down apparent performance

Chip Packaging



(a) 8 Mbit EPROM



(b) 16 Mbit DRAM

1Mbit x 8: $2^{20} = 1\text{M}$: A0 – A19

Data lines: D0 – D8 (8 bit words)

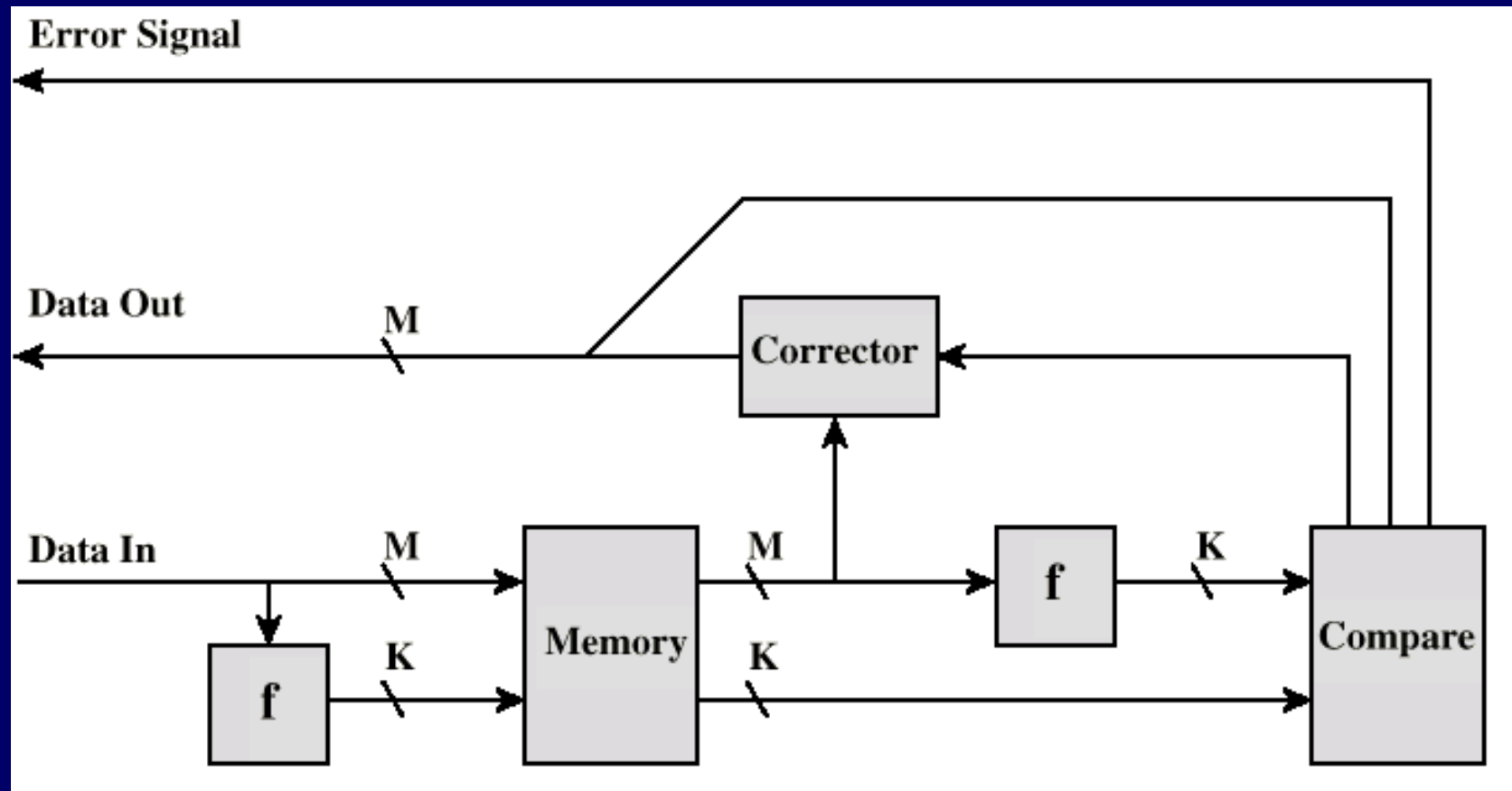
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Error Correction

- Hard Failure
 - Permanent physical defect
 - Caused by harsh environmental abuse, wear, etc.
- Soft Error
 - Random, non-destructive
 - No permanent damage to memory
 - Caused by power supply problems, alpha particles, etc.
- Memory needs logic for detecting and correcting errors

Error Correcting Code Function



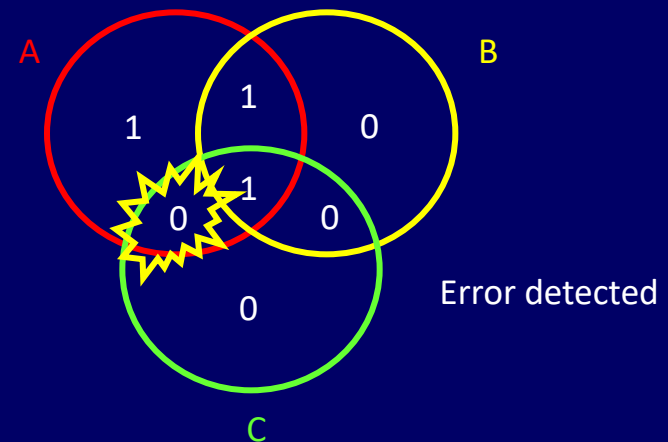
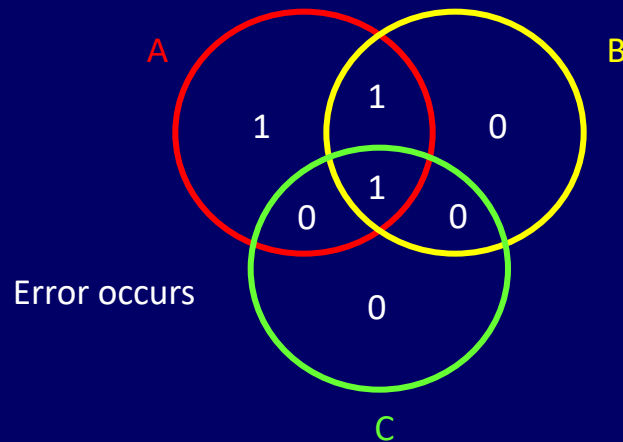
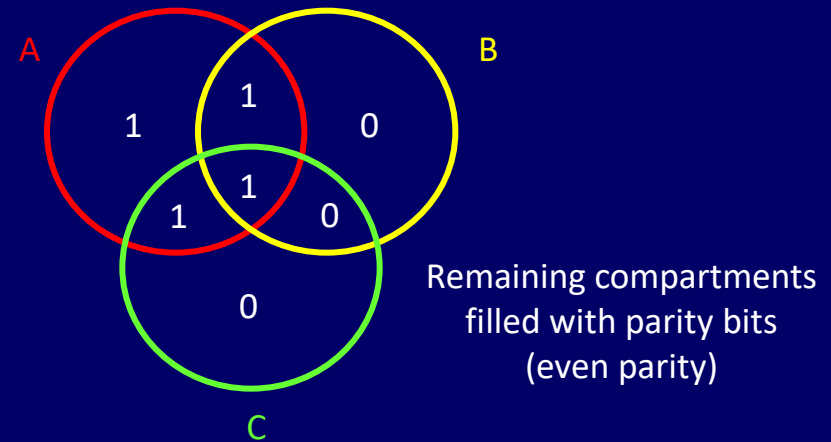
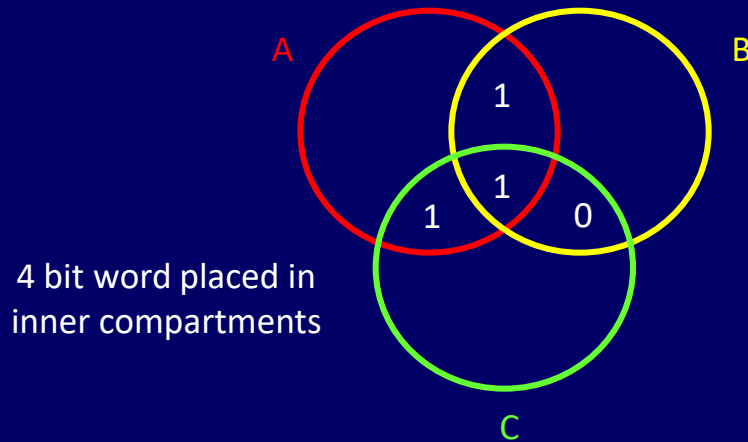
Error Correcting Code Function

- When data written into memory
 - Calculation function, f , is performed and produces a code
 - Both data (M bits) and code (K bits) are stored in memory
- When data read out
 - Data (M bits) used to generate a new code (K bits) using same function, f
 - Both codes (before memory write and after memory read) are compared

Error Correcting Code Function

- Three results
 - No errors
 - Error detected
 - Possible to correct by feeding data bits and error correction bits into a corrector
 - Error detected
 - Not possible to correct
 - Report error
- Codes = error correcting codes
- Simplest error correcting codes
 - Hamming error correcting code

Hamming Error Correcting Code



Hamming Error Correcting Code

- Comparison logic receives 2 K-bit values (check bits) as input
 - Both values are XOR-ed = syndrome word
 - Syndrome word range = 0 to $2^K - 1$
 - If 0 = no errors, else value indicates which bit is in error
- Error may be in M data bits, or K check bits
 - So, $2^K - 1 \geq M + K$
 - E.g. for word with 8 data bits ($M = 8$), must have 4 check bits ($K = 4$)
 - $K = 4$: $2^4 - 1 > 8 + 4$ (sufficient)
 - $K = 3$: $2^3 - 1 < 8 + 3$ (not sufficient)

Hamming Error Correcting Code

- Characteristics
 - If syndrome word (XOR-ed check bits) all 0s, no error
 - If syndrome word has only one bit set to 1, then error occurred to one of the check bits.
 - No correction needed
 - If syndrome word has more than one bit set to 1, then value of syndrome word indicates position of the data bit in error
 - Correction done by inverting that bit location

Hamming Error Correcting Code

- Example for 8-bit data word with 4 check bits (M = 8, K = 4)

Bit position	12	11	10	9	8	7	6	5	4	3	2	1
Data / Check bits	1100	1011	1010	1001	1000	0111	0110	0101	0100	0011	0010	0001
	D8	D7	D6	D5	C8	D4	D3	D2	C4	D1	C2	C1

- Check bits calculated as follows:

$$\begin{aligned}
 \circ C1 &= D1 \oplus D2 \oplus D4 \oplus D5 \oplus D7 \\
 \circ C2 &= D1 \oplus D3 \oplus D4 \oplus D6 \oplus D7 \\
 \circ C4 &= D2 \oplus D3 \oplus D4 \oplus D8 \\
 \circ C8 &= D5 \oplus D6 \oplus D7 \oplus D8
 \end{aligned}$$

Hamming Error Correcting Code

- Example 8-bit data word = 00111001
- Check bits calculated as follows:

$$\begin{aligned}
 \circ C1 &= D1 \oplus D2 \oplus D4 \oplus D5 \oplus D7 = 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 = 1 \\
 \circ C2 &= D1 \oplus D3 \oplus D4 \oplus D6 \oplus D7 = 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 = 1 \\
 \circ C4 &= D2 \oplus D3 \oplus D4 \oplus D8 = 0 \oplus 0 \oplus 1 \oplus 0 = 1 \\
 \circ C8 &= D5 \oplus D6 \oplus D7 \oplus D8 = 1 \oplus 1 \oplus 0 \oplus 0 = 0
 \end{aligned}$$

Bit position	12	11	10	9	8	7	6	5	4	3	2	1
Data / Check bits	1100	1011	1010	1001	1000	0111	0110	0101	0100	0011	0010	0001
Stored	D8	D7	D6	D5	C8	D4	D3	D2	C4	D1	C2	C1
	0	0	1	1	0	1	0	0	1	1	1	1

Hamming Error Correcting Code

- Example 8-bit data word fetched = 00111**1**01

Bit position	12	11	10	9	8	7	6	5	4	3	2	1
Data / Check bits	1100	1011	1010	1001	1000	0111	0110	0101	0100	0011	0010	0001
Stored	D8	D7	D6	D5	C8	D4	D3	D2	C4	D1	C2	C1
Stored	0	0	1	1	0	1	0	0	1	1	1	1
Fetched	0	0	1	1	0	1	1	0	1	1	1	1

- Check bits re-calculated as follows:

$$\begin{aligned}
 \circ C1 &= D1 \oplus D2 \oplus D4 \oplus D5 \oplus D7 = 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 = 1 \\
 \circ C2 &= D1 \oplus D3 \oplus D4 \oplus D6 \oplus D7 = 1 \oplus 1 \oplus 1 \oplus 1 \oplus 0 = 0 \\
 \circ C4 &= D2 \oplus D3 \oplus D4 \oplus D8 = 0 \oplus 1 \oplus 1 \oplus 0 = 0 \\
 \circ C8 &= D5 \oplus D6 \oplus D7 \oplus D8 = 1 \oplus 1 \oplus 0 \oplus 0 = 0
 \end{aligned}$$

Hamming Error Correcting Code

- Example 8-bit data word fetched = 00111101
- New check bits compared with old check bits

	C8	C4	C2	C1	
	0	1	1	1	(Old)
$\oplus$	0	0	0	1	(New)
	0	1	1	0	

- Check bits 0110 indicates bit position 6 or data bit 3 (D3) is in error

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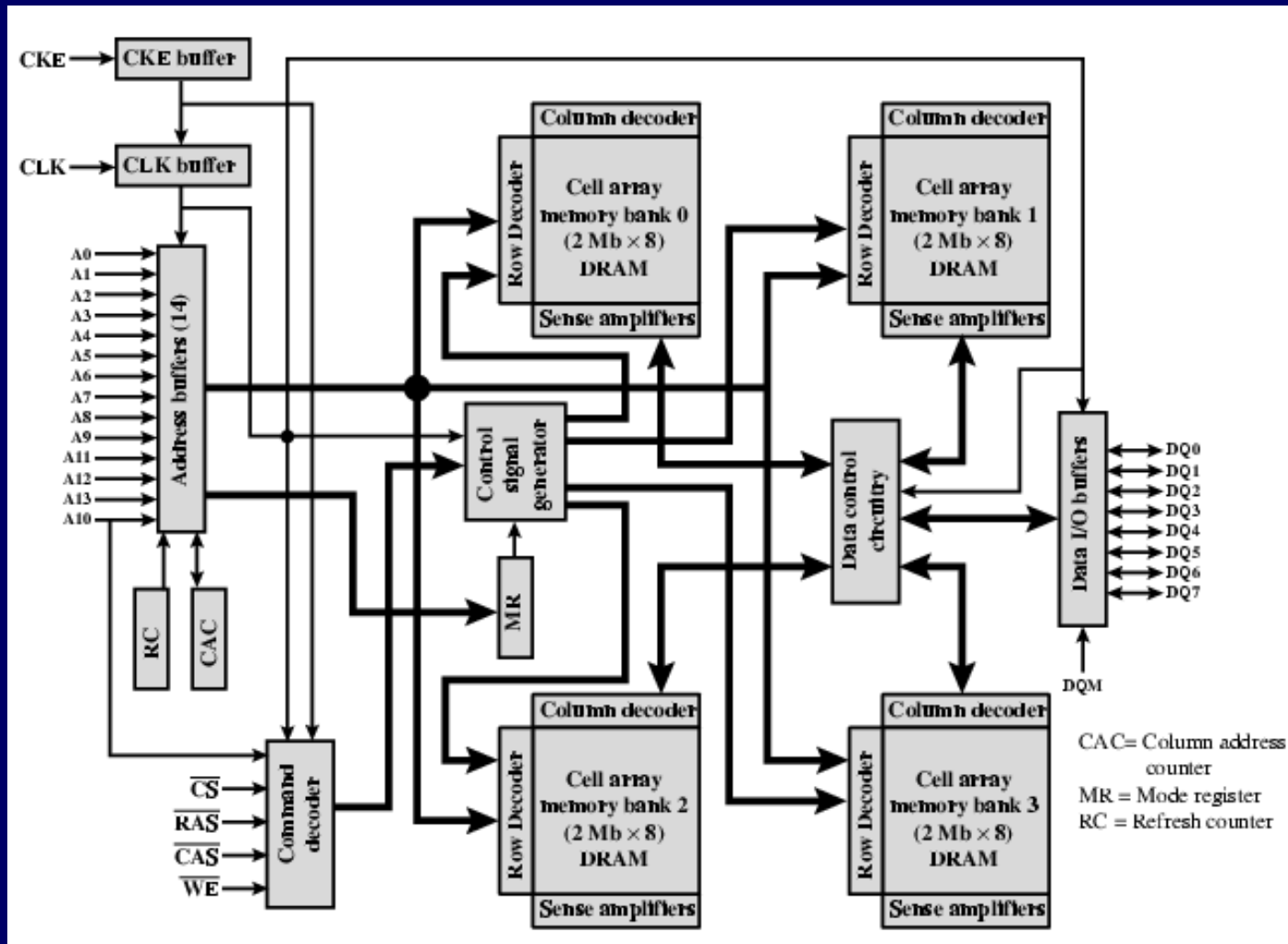
Advanced DRAM Organization

- Basic DRAM same since first RAM chips
- Enhanced DRAM
 - Contains small SRAM as well
 - SRAM holds last line read (c.f. Cache!)
- Cache DRAM
 - Larger SRAM component
 - Use as cache or serial buffer

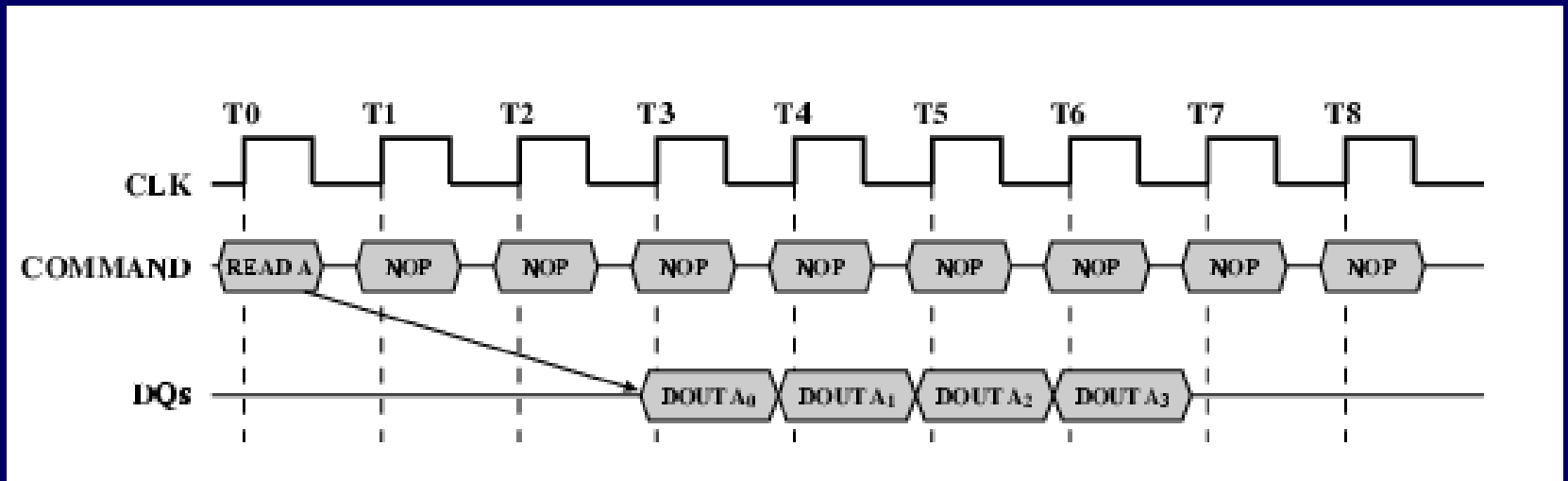
Synchronous DRAM (SDRAM)

- Access is synchronized with an external clock
- Address is presented to RAM
- RAM finds data (CPU waits in conventional DRAM)
- Since SDRAM moves data in time with system clock, CPU knows when data will be ready
 - CPU does not have to wait, it can do something else
- Burst mode allows SDRAM to set up stream of data and fire it out in block
- DDR-SDRAM sends data twice per clock cycle (leading & trailing edge)

IBM 64Mb SDRAM



SDRAM Operation



Burst length = number of separate units of data synchronously fed onto bus
= 4

Latency = time between receipt of read request and beginning of data transfer
= 2

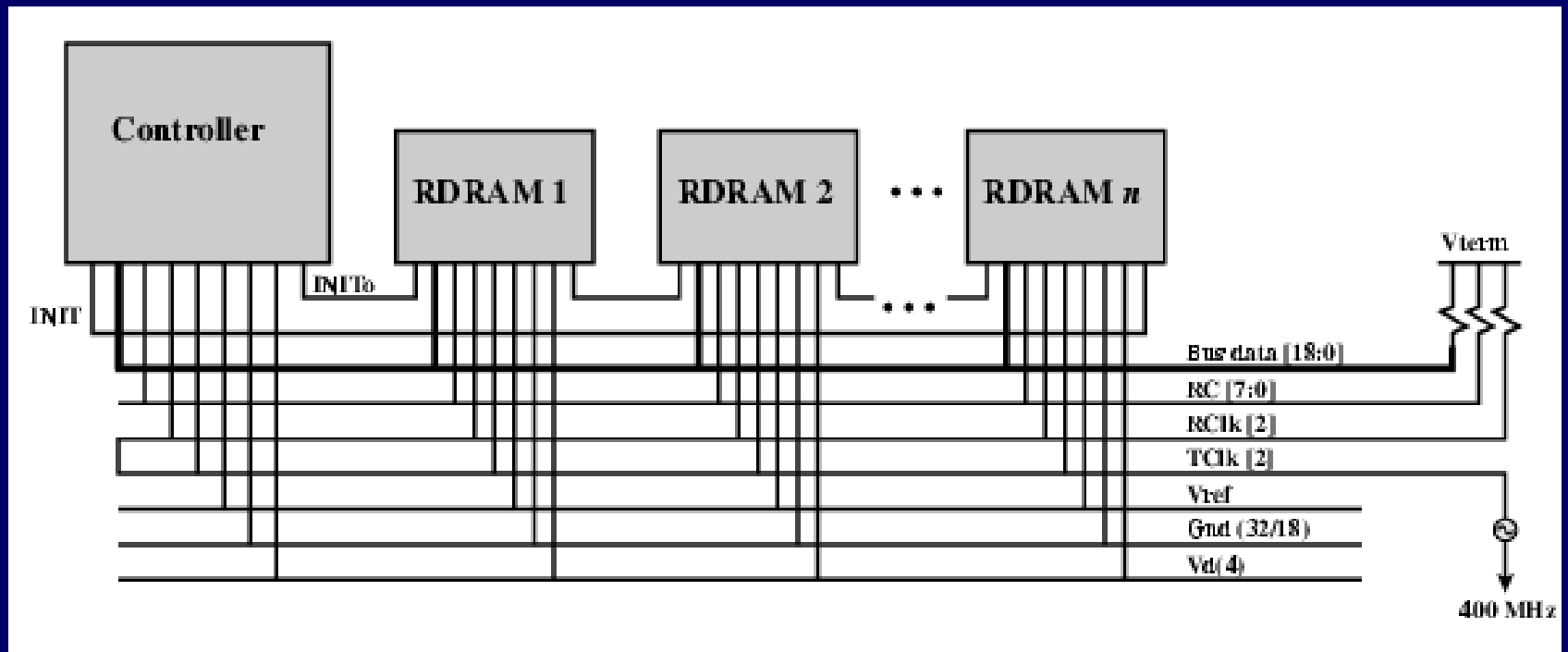
RAMBUS DRAM

- Adopted by Intel for Pentium & Itanium
- Main competitor to SDRAM
- Vertical package (all pins on one side)
- Data exchange over 28 wires < 12 cm long
- Bus addresses up to 320 RDRAM chips at 1.6Gbps
- Asynchronous block protocol
 - 480ns access time
 - Then 1.6 Gbps data rate

RAMBUS DRAM

- Rambus speed possible because of bus itself
 - Defines impedances, clocking and signals very precisely
 - Does not require PAS, CAS, R/W and CE signals
 - Gets memory request over high-speed bus
 - Contains address, operation type and number of bytes

RAMBUS Diagram



Cache DRAM

- Developed by Mitsubishi
- Integrates small SRAM cache (16KB) onto generic DRAM chip
- SRAM on the CDRAM used in two ways
 - As a true cache (for ordinary random access)
 - As a buffer (for serial access of a block of data)